

Problem 8

Problem 8

**Problem 1** The function  $f$  is defined by  $f(x) = \frac{\sqrt{2x^2 + 1}}{3x - 5}$ .

- (a) Find all vertical asymptotes of  $f$ . **EXPLAIN** and justify your answer by using appropriate limits.
- (b) Find all horizontal asymptotes of  $f$ . **EXPLAIN** and justify your answer by using appropriate limits..